

Stage 1 Protocol inputs

Manifest and split

PLC-BEAD accounting

2,431 binaries
10-fold inferred-name split

Read discipline

Allowlisted paths
SHA-256 + fold assertion

Fabrication guard

XOR

Fixture / dataset modes
mutually exclusive

Stage 2 Observations

Tool output and gaps

nm-readable branch

1,876 non-CODESYS rows
symbol + rule provenance

No-symbol branch

555 CODESYS .app rows
U01 tool boundary

Typed graph records

typed nodes and edges
provenance links

Three-valued evidence-card fill: 1 = supported, 0.5 = bounded/uncertain, 0 = missing without explanatory event

Stage 3 Evidence contract

Support and uncertainty

EC1 Function candidate	EC2 Core/control hint	EC3 Static facts
EC4 Declared POU role	EC5 Relation graph	EC6 Uncertainty coverage

Stage 4 Validation

Oracle, invariants, audit

Uncertainty audit

U01--U14 + field mapping

Execution audit

2,431 hash reads; 0 mismatch

Validity checks

source oracle + 5 invariants

Scope: evidence traceability on PLC-BEAD and one bounded second pipeline;
not runtime-state recovery, semantic accuracy, architecture generalization, or vulnerability detection.